Experiment 6 Day 3: Conversion of Borneol to Camphor

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Introduction

In this experiment, borneol was converted to camphor using dichloromethane, acetic acid, sodium bicarbonate, and sodium hypochlorite. The organic product was collected via reflux, and the percent yield was calculated.

Materials and Methods

In a 10 mL long-neck round-bottom flask, 0.3 g of the racemic borneol mixture product and 0.3 mL of glacial acetic acid was added. The mixture was then dissolved in 1.0 mL of acetone, and a boiling chip was added to the flask. A simple distillation apparatus was assembled, and a water bath between 40-50℃ was included. Once all of the reagents were dissolved, 6.0 mL of aqueous sodium hypochlorite solution was added to the mixture. The addition was done in a dropwise fashion over a 35 minute increment of time. After each addition of the sodium hypochlorite, the flask was gently swirled as to ensure thorough mixing of the reagents. The reaction was allowed to continue for 15 minutes until it was complete. After completion, the mixture was cooled to room temperature. The camphor was extracted with two portions (2.0 mL each) of dichloromethane. A centrifuge tube was used to make the extraction more efficient. Once the extraction was complete, the combined methylene chloride layers were neutralized with approximately 2.0 mL of saturated sodium bicarbonate. The addition of the saturated sodium bicarbonate stopped once there was cessation of carbon dioxide bubbled. After the mixture was neutralized, the organic solution was washed sequentially with 2.0 mL of 5% sodium bisulfite and with 2.0 mL of water. The organic solution was dried with anhydrous sodium sulfate. Finally, the dichloromethane was evaporated over a water bath with the temperature not exceeding 50℃. The collected crystals were weighed, and the percent yield was calculated.

Results

Weight of Organic Product (Camphor)	0.183 grams

Table 1. Weight of final product collected.

Calculations

In this experiment, 0.183 g of organic product (camphor) was collected from an initial 0.3 g of borneol. The percent yield and theoretical yield calculations are as follows:

Theoretical Yield:

$$(0.3\ g\ borneol)x(154.25\ g/mol)\ =\ 0.0019\ mol\ borneol$$

$$(0.183\ g\ camphor)x(152.24\ g/mol)\ =\ 0.0012\ mol\ camphor$$

$$(0.0012\ mol\ camphor)x(154.25\ g/mol)\ =\ 0.185\ g\ camphor$$

Percent Yield:

Actual Yield = 0.183 g camphor

$$(0.183)/(0.185\ x\ 100)\ =\ 98.9\%$$

Discussion

In this experiment, borneol was converted to camphor. Camphor appears as white, waxy solid with a strong earthy odor. These characteristics were observed after the completion of this experiment. Possible sources of error include not allowing the reaction to proceed for long enough and the starting material being impure. Both of these could lead to a reduced yield of organic product. Not allowing the reaction to proceed for long enough can result in by-product formation or an insufficient conversion of borneol to camphor. One way to fix this would be to allow the reaction to run for a longer amount of time. Also, before removing the flask from the water bath and reflux apparatus, ensure that two distinct layers are present in the mixture. In terms of the starting material being impure, there could have been trace amounts of water and

isobornyl acetate. To fix this, make sure there are free rolling crystals of anhydrous sodium sulfate in the solution before proceeding. This ensures all microscopic amounts of water are removed from the organic product. Furthermore, to ensure all of the isobornyl acetate is removed from the borneol sample, the reaction can run for a longer amount of allotted time in Day 2 of the experiment to increase the chances that all of the isobornyl acetate is converted to borneol. The calculated theoretical yield for this experiment is 0.185 g of camphor; the observed yield is 0.183 g of camphor. The percent yield is 98.9%. Although the percent yield is high, meaning the majority of the borneol used was converted to camphor, it is still likely that sources of error occurred.

References

Keller L, Joseph K, Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.